Area Between Curves

ANSWER KEY

Area between two curves

1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$.

Intersections: $x^2 = x + 2$, so $x^2 - x - 2 = 0$, $(x-2)(x+1) = 0$: $x = -1, 2$. Area $= \int_{-1}^{2} [(x+2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \frac{9}{2}$.

2 The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.

$\int_0^4 \sqrt{x}\, dx = \left[\frac{2}{3}x^{3/2}\right]_0^4 = \frac{16}{3}$.

Setting up with respect to y, and total area

3 Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y.

Intersections: $y^2 = 2y$ gives $y = 0, 2$. Area $= \int_0^2 (2y - y^2)\, dy = \left[y^2 - \frac{y^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

4 Find the total area between $f(x) = \sin x$ and the x-axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).

By symmetry, $\int_0^{\pi} \sin x\, dx = 2$ and $\left|\int_{\pi}^{2\pi} \sin x\, dx\right| = 2$. Total area $= 2 + 2 = 4$.

More area problems

5 Find the area of the region bounded by $y = x^2 - 4$ and $y = 4 - x^2$.

Intersections: $x^2 - 4 = 4 - x^2$, so $2x^2 = 8$, $x = \pm 2$. Upper curve is $y = 4 - x^2$: Area $= \int_{-2}^{2} [(4 - x^2) - (x^2 - 4)] dx = \int_{-2}^{2} (8 - 2x^2)\, dx = \left[8x - \frac{2x^3}{3}\right]_{-2}^{2} = \frac{64}{3}$.

6 Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).

Area $= \int_0^1 (x - x^3)\, dx = \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

Areas requiring case splits

7 Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).

On $[-1, 0]$, $x^3 \geq x$; on $[0, 1]$, $x \geq x^3$. Area $= \int_{-1}^{0} (x^3 - x)\, dx + \int_0^1 (x - x^3)\, dx$. By symmetry both integrals equal $\frac{1}{4}$ (using the earlier result), so total area $= \frac{1}{2}$.

8 Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.

Intersections: $2x = x^2$ gives $x = 0, 2$. On $(0, 2)$, $2x > x^2$: Area $= \int_0^2 (2x - x^2)\, dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

Areas involving trigonometric and exponential curves

9 Find the area between $y = \cos x$ and $y = 0$ on $\left[0, \frac{\pi}{2}\right]$.
Area $= \int_0^{\pi/2} \cos x \, dx = [\sin x]_0^{\pi/2} = 1 - 0 = 1$.

10 Find the area between $y = e^x$ and $y = e^{-x}$ on $[0, 1]$ (note $e^x \geq e^{-x}$ on this interval).
Area $= \int_0^1 (e^x - e^{-x}) \, dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + \frac{1}{e} - 2$.

Visualizing a region bounded by two curves

11 The graphs of $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$ from an earlier problem are shown together. Confirm from the picture that the region is symmetric about the y-axis, and use this to verify the earlier area answer $\frac{64}{3}$ using $2\int_0^2 (8 - 2x^2) \, dx$.

The region is symmetric about $x = 0$ since both curves are even functions.
$2\int_0^2 (8 - 2x^2) \, dx = 2\left[8x - \frac{2x^3}{3}\right]_0^2 = 2\left(16 - \frac{16}{3}\right) = 2 \cdot \frac{32}{3} = \frac{64}{3}$, matching the earlier full-interval computation.

Area between a curve and a horizontal line

12 Find the area between $y = 6 - x^2$ and $y = 2$.

Intersections: $6 - x^2 = 2$ gives $x = \pm 2$. Area $= \int_{-2}^{2} [(6 - x^2) - 2] dx = \int_{-2}^{2} (4 - x^2) \, dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = 2\left(8 - \frac{8}{3}\right) = \frac{32}{3}$.

13 Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ for $x \geq 0$.

Intersections: $\sqrt{x} = \frac{x}{2}$ gives $x = 0$ or $x = 4$. On $(0, 4)$, $\sqrt{x} \geq \frac{x}{2}$: Area $= \int_0^4 \left(\sqrt{x} - \frac{x}{2}\right) dx = \left[\frac{2}{3}x^{3/2} - \frac{x^2}{4}\right]_0^4 = \frac{16}{3} - 4 = \frac{4}{3}$.

Area between curves in applied contexts

14 Two runners' velocities (m/s) over a 10-second race are $v_1(t) = 2t$ and $v_2(t) = t + 3$. The area between these curves represents the gap in distance traveled. Find this gap over $[0, 10]$.

Intersection: $2t = t + 3$ gives $t = 3$. For $t < 3$, $v_2 > v_1$; for $t > 3$, $v_1 > v_2$. Gap $= \left|\int_0^3 (v_2 - v_1) dt\right| + \left|\int_3^{10} (v_1 - v_2) dt\right|$. $\int_0^3 (3 - t) \, dt = \left[3t - \frac{t^2}{2}\right]_0^3 = 9 - 4.5 = 4.5$.
$\int_3^{10} (t - 3) \, dt = \left[\frac{t^2}{2} - 3t\right]_3^{10} = (50 - 30) - (4.5 - 9) = 20 + 4.5 = 24.5$. Total gap $= 4.5 + 24.5 = 29$ m.

15 Find the area of the region bounded by $y = x^2 - 2x$ and the x-axis on $[0, 2]$ (note the curve dips below the axis on this interval).
$y = x^2 - 2x = x(x - 2)$ is ≤ 0 on $[0, 2]$. Area $= \left|\int_0^2 (x^2 - 2x) \, dx\right| = \left|\left[\frac{x^3}{3} - x^2\right]_0^2\right| = \left|\frac{8}{3} - 4\right| = \frac{4}{3}$.

16 Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$, splitting at the roots $x = -1, 0, 1$ where the sign changes.

On $[-1, 0]$, $x^3 - x \geq 0$; on $[0, 1]$, $x^3 - x \leq 0$. Area $= \int_{-1}^{0} (x^3 - x)\,dx + \left| \int_0^1 (x^3 - x)\,dx \right| = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} + \left| \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1 \right| = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.